Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	16
interior lacuna count	16
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 16)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1420.45	1293	702.329	px^2
major_axis_length_px	lacuna shape	64.3118	54.0186	22.3213	px
minor_axis_length_px	lacuna shape	35.2082	30.951	7.6188	px
aspect_ratio	lacuna shape	1.8719	1.7017	0.7153	unitless
eccentricity	lacuna shape	0.7783	0.8091	0.1414	unitless
solidity	lacuna shape	0.7237	0.7085	0.0904	unitless
roots_count	headline	8.125	9	4.8973	count
ring_length_r30_px	headline	310.25	315	176.394	px
ring_length_r60_px	reported	575.375	605.5	355.606	px
owned_length_px	ownership-dependent	882.505	715.47	1080.32	px
edge_count	ownership-dependent	48.75	32.5	61.9737	count
mean_edge_length_px	ownership-dependent	17.4621	17.7052	5.4589	px
perimeter_px	normalised	219.641	184.48	100.453	px
ring_area_r30_px2	normalised	6570.25	7072	2510.45	px^2
ring_area_r60_px2	normalised	15758	18338.5	6991.56	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.044879	0.0466742	0.0157578	px^-1
ring_density_r60	normalised	0.0348686	0.0342042	0.0130523	px^-1
roots_per_100px_perimeter	normalised	3.6201	3.677	1.8875	per 100 px
ring_attached_length_r30_px	network v2	277.125	269.5	178.891	px
ring_attached_length_r60_px	network v2	473.25	490.5	350.03	px
ring_length_w_r30_px	network v2	350.62	359.028	198.119	px
ring_length_w_r60_px	network v2	646.368	681.408	397.66	px
sholl_crossings_r10	network v2	7.125	7	4.0311	count
sholl_crossings_r20	network v2	7.9375	8	4.5235	count
sholl_crossings_r30	network v2	7.75	7	4.7958	count
loop_count	loops, report only	3.0625	0	6.0274	count
loop_edge_share	loops, report only	0.1147	0	0.1547	share
ring_width_mean_r30_px	width	7.6903	7.635	1.2948	px

Per field
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measure	value	units
analysed_area_px2	869119	px^2
total_skeleton_length_px	19682	px
canalicular_length_density_per_px	0.0226459	px^-1
junction_count	421	count
junction_density_per_px2	0.0004844	px^-2
lacuna_count	16	count

lacunae_per_px2	1.841e-05	px^-2
skeleton_component_count	295	count
mean_component_length_px	66.7186	px
median_component_length_px	16	px
field_length_density_w_per_px	0.025267	px^-1
field_density_without_flagged_per_px	0.0171984	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	577	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0226459	px^-1
field_length_density_with_unattached_per_px	0.0233098	px^-1
excluded_area_px2	161365	px^2
analysed_field_area_px2	887211	px^2
loop_count	56	count
loop_edge_share	0.1756	share
edge_count	1281	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.4474	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	16640	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
2	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
3	933	52.1712	30.6068	1.7046	0.8098	0.7272	-0.9318	146.222	325.69	False
4	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
5	3093	92.0714	54.1062	1.7017	0.8091	0.6916	1.2856	410.259	393.304	False
6	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
7	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
8	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
9	634	48.158	30.1352	1.5981	0.78	0.5992	-1.5103	631.199	551.279	False
10	1329	62.4213	28.6247	2.1807	0.8887	0.8819	1.2484	643.479	606.14	False
11	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
12								290.753	563.455	False
13								291.903	675.169	False
14								565.978	720.266	False
15								614.615	620.067	False
16								698.601	352.387	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	377	731	915.181	53	17.2676	8.4862

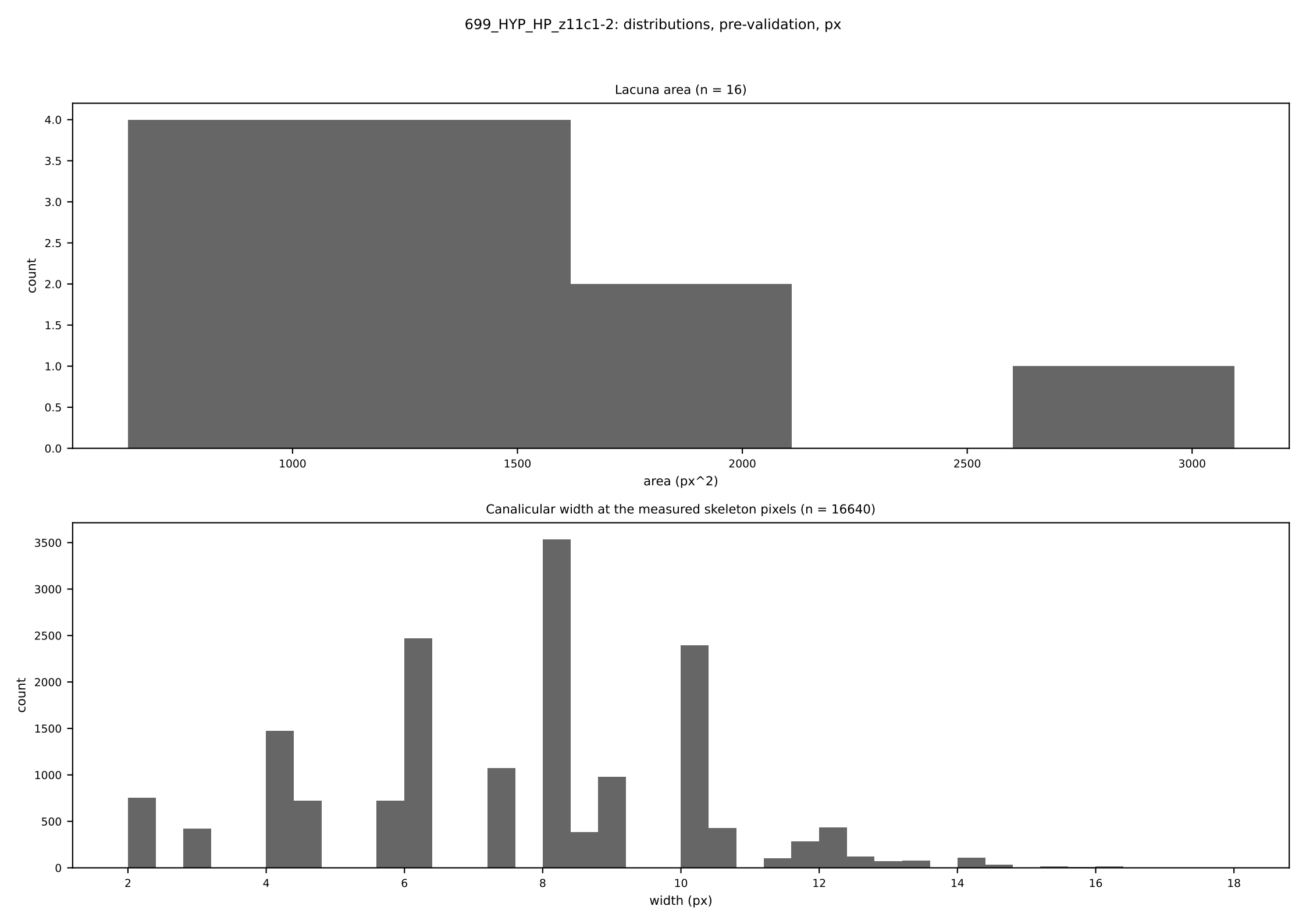
2	12	386	693	717.156	30	23.9052	8.1517
3	8	302	657	1005.88	59	17.0488	9.3181
4	2	209	470	315.622	15	21.0415	6.8106
5	18	813	1542	4371.17	252	17.3459	7.4887
6	10	328	433	468.848	20	23.4424	7.7813
7	2	118	232	17.8995	1	17.8995	4.7624
8	13	424	731	728.813	35	20.8232	7.4846
9	10	204	283	350.22	20	17.511	6.2316
10	5	214	302	216.48	12	18.04	7.231
11	10	360	625	789.713	37	21.3436	8.5944
12	11	404	860	1031.68	65	15.8721	8.2415
13	2	228	586	713.784	46	15.517	7.3283
14	0	10	10	0	0	0	10.3389
15	8	210	229	187.752	14	13.4109	6.5505
16	7	377	822	2289.87	121	18.9246	8.2448

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.622	7458	18845	1	1	0.0505497	0.0387901	6.2623
2	234.25	8068	19848	1	1	0.0478433	0.0349154	5.1227
3	172.995	7102	19616	1	1	0.0425232	0.0334931	4.6244
4	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
5	467.446	11796	28237	1	1	0.0689217	0.0546092	3.8507
6	318.333	7208	13815	1	1	0.045505	0.0313427	3.1414
7	232.285	5734	14497	1	1	0.020579	0.0160033	0.861
8	409.931	10384	22845	1	1	0.0408321	0.0319982	3.1713
9	174.853	4702	8707	1	1	0.0433858	0.0325026	5.7191
10	172.267	3574	7974	1	1	0.0598769	0.0378731	2.9025
11	182.894	6958	19375	1	1	0.051739	0.0322581	5.4676
12	186.066	7155	17832	1	1	0.056464	0.0482279	5.9119
13	91.9828	5369	14348	1	1	0.042466	0.0408419	2.1743
14	104.468	1694	2277	1	1	0.00590319	0.00439174	0
15	228.35	3529	4140	1	1	0.0595069	0.055314	3.5034
16	172.439	7042	19502	1	1	0.0535359	0.0421495	4.0594

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	348	593	432.99	829.039	8	8	13
2	374	618	434.049	765.115	9	12	12
3	272	599	349.907	743.428	6	9	7
4	169	284	233.752	529.76	4	4	7
5	799	1489	916.37	1721.76	18	18	17
6	267	334	368.149	489.233	9	9	6
7	47	47	128.983	252.823	3	4	4
8	400	491	468.605	820.754	11	13	7
9	190	207	230.581	316.936	5	4	3
10	195	242	251.108	344.877	6	5	2
11	321	490	406.563	701.6	8	11	9
12	325	657	450.463	964.21	8	11	13
13	159	521	261.865	661.215	3	8	8
14	10	10	12.0711	12.0711	1	0	0
15	210	229	239.066	260.137	5	3	3
16	348	761	425.392	928.938	10	8	13


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 16)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 16640)
count
width (px)